

JAIDEEP NUTALAPATI

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EDUCATION

- **UNIVERSITY OF ILLINOIS AT CHICAGO** August 2024 – May 2026
Master of Computer Science Chicago, IL
- **JAIN UNIVERSITY** August 2020 – May 2024
Bachelor of Technology in Computer Science and Engineering Bangalore, India

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL (MySQL, PostgreSQL, SQLite)
- **ML & Statistics:** Supervised & Unsupervised Learning, Regression, Classification, Random Forest, XGBoost, KNN, SVD++, SVM, K-means, Feature Engineering, Hyperparameter Tuning, Cross Validation, Model Evaluation, Predictive Modeling, Statistical Analysis, A/B Testing, Hypothesis Testing, Data Analytics & Visualization.
- **Frameworks & Libraries:** Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Plotly, FastAPI, Flask, Evidently AI.
- **Tools:** Docker, Git, GitHub, Power BI, Tableau, Snowflake, Streamlit, Prometheus, Grafana, Postman

EXPERIENCE

UIC Campus Housing April 2025 – May 2026
Building Service Aide — Data & Operations Analyst Chicago, IL

- Analyzed 25,000+ furniture asset records collected across 9 UIC housing buildings and 1,000+ apartments using Google Sheets and Python to evaluate asset condition and replacement needs.
- Performed EDA and developed condition-based visualizations to classify assets into three condition tiers and identify replacement priorities.
- Delivered a data-driven stakeholder report and visual findings to housing management, supporting a \$47–50M asset replacement budget estimate across UIC residential buildings.
- Collaborated with a 7-member cross-functional team across Summer and Winter 2025 audit cycles, maintaining data accuracy and consistency across 25,000+ records.

Department of Pediatrics, UIC February 2026 – May 2026
Web Developer Graduate Hourly Chicago, IL

- Analyzed website performance using Google Analytics (GA4) across all departmental pages, monitoring monthly page views and 38,000+ user interactions and presenting data-driven findings to department leadership.
- Redesigned and maintained all departmental web pages using HTML, JavaScript, and WordPress ensuring compliance with ADA accessibility standards and federal web guidelines, and optimizing front-end performance by standardizing component architecture to improve page load time and consistency.

PROJECTS

Race Pulse – Marathon Finish Time Prediction

- Built and evaluated supervised machine learning models including Random Forest and XGBoost on 79,000+ race records integrated with real-time weather data, achieving R^2 of 0.99 through feature engineering, hyperparameter tuning, and cross-validation.
- Performed end-to-end ML pipeline development including data wrangling, feature selection, model training, and evaluation using RMSE and R^2 metrics, and communicated predictive insights through interactive Plotly and Seaborn visualizations.

Movie Recommendation System

- Developed a recommendation engine using SVD++ and KNN collaborative filtering, achieving an RMSE of 0.58 and deploying the model as a REST API using FastAPI for real-time inference.
- Implemented CI/CD using Docker and Jenkins and monitored deployed model performance and data drift using Prometheus, Grafana, Evidently AI, and Pandera.

NYC Street Tree Census – Urban Data Analytics

- Processed and cleaned 680,000+ tree records from the 2015 NYC Street Tree Census, handling missing values, standardizing categorical fields, and generating feature embeddings using species, health, and borough attributes.
- Applied PCA and UMAP for dimensionality reduction and developed an interactive linked dashboard with scatterplot, treemap, and map views, enabling brushing and tooltip interactions to explore geographic and health-based patterns.

Hotel Management & Booking Analytics System

- Designed and deployed a normalized PostgreSQL database with 10 relational tables using primary/foreign keys, composite keys, constraints, and relationship modeling to maintain data integrity across hotel, booking, customer, room, and review data.
- Developed analytical SQL queries using JOINS, GROUP BY, HAVING, subqueries, EXISTS/NOT EXISTS, and aggregate functions to analyze booking volume, top clients, hotel ratings, customer spending, room utilization, and geographic booking patterns; integrated queries into a Python/Flask application using psycopg2.